UNIVERSITI MALAYSIA SARAWAK

Faculty of Computer Science and Information Technology

TMA3084 Software Engineering Laboratory | Semester 1, 2026/2027

PROJECT PROPOSAL

(Product Requirements Document)

<Project Title (OutageWatchSarawak)>

Group Number / Name	*Team 24 - Biscuding*
Lab Group / Demonstrator	*Group 2 – Raynold Kabai*
Client / Organisation	*Clay Anak Ranyun, position, Sarawak Energy Berhad*
Lecturer	Mdm. Nurfauza Jali
GitHub Repository	*https://github.com/104106/TMA3084_G02_T24_OutageWatchSarawak*
Version / Date	*<v1.0 (original) | v1.1 (revised after presentation) – dd/mm/yyyy>*

Team Members

No.	Name	Matric No.	TSPi Role	Signature
1	Dayrill Muller Anak Clay	103837	Team Leader + Client Liaison	
2	Fabian Anak Sudin	104106	Quality/Process Manager + Support Manager	
3	Maccoy Bong Nai Sean	104822	Planning Manager	
4	Bricolen Benjamin	98371	Development Manager	

Declaration

By signing above, we certify that this proposal is our own work, that all external sources, libraries, APIs and AI-assisted content are acknowledged, and that we have not engaged in plagiarism or unauthorised collusion (Peraturan Akademik UNIMAS, Etika Akademik). AI tool usage is recorded in LOGAI.

How to Use This Template (delete this page before submission)

‣ Grey italic text (‣) and blue 'PRD section' bars are guidance. Replace them with your own content, then delete them.

‣ Target length: 10–15 pages excluding cover and appendices. Be concise: tables beat long paragraphs.

‣ Formatting: Times New Roman 12 pt, 1.5 line spacing, left aligned. Heading 1 = 14 pt bold, Heading 2 = 12 pt bold. Right-click the Table of Contents → Update Field before submitting.

‣ Every figure and table needs a numbered caption and must be referred to in the text. Add alt text to images.

‣ Keep IDs consistent: the FR-xx IDs you define here are reused in the SRS, GitHub Issues, test cases (LOGTEST) and LOGAI.

Proposal Checklist → PRD Mapping

Proposal checklist item	PRD equivalent	Key question it answers
Abstract	Executive Summary / TL;DR	What are we building, for whom, and why?
Introduction	Background & Context	Who is the client and who is the team?
Problem Statement	Problem & Opportunity	What pain exists today and what is the evidence?
Project Objectives	Goals & Success Metrics	How will we know the project succeeded?
Project Perspective	Users/Personas & System Context	Who uses it and how does it fit with other systems?
Project Scope	Scope, Features & Requirements	What is in, what is out, and what is the AI feature?
Task Allocation + Team Contribution	Team / Stakeholders (RACI)	Who owns what?
Team Goals	Team Working Agreement	How will the team work and measure itself?
Expected Result	Release Criteria / Deliverables	What will be handed over at the end?
Project Planning	Milestones, Timeline, Risks	When, in what order, and what could go wrong?

Table A: Mapping of proposal checklist to PRD sections

‣ Simplification: 'Task Allocation' and 'Team Contribution' are merged into one table; 'Software Methodology' and 'System Environment' are folded into Sections 9 and 5; placeholder writing tips from the old template were removed.

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▸ Right-click → Update Field to refresh page numbers. Add a List of Figures / List of Tables if you have more than three of each.

Abstract

PRD section: Executive Summary / TL;DR **Rubric link:** Clarity & presentation

▸ *One paragraph, 150–250 words. Write it LAST. Cover: client and context → problem → proposed Flutter mobile app → the AI feature → main expected outcome.*

1. Introduction

PRD section: Background & Context **Rubric link:** Background (Section A)

Power outages affect households and businesses across Sarawak, and fast, accurate reporting helps utility teams respond sooner. Currently, customers report faults through [phone/other channels, confirm with client], which can lead to incomplete location details and little feedback about progress. OutageWatch Sarawak is an Android prototype that lets customers report an outage with a GPS location and photo, uses AI to suggest the fault category, and lets customers track the status of their report. A mobile app suits this problem because outages happen anywhere, and most customers already carry a smartphone with GPS and a camera.

1.1 Client Background

▸ *Organisation: Sarawak Energy Berhad (SEB), the main electricity utility in Sarawak.*

1.2 Team Background

▸ *Team Biscuding, motto ' Du bist gut genug'. Our members have experience in Flutter.*

2. Problem Statement

PRD section: Problem & Opportunity **Rubric link:** Objectives / constraints analysed

▸ *Describe how the client works today and what goes wrong. Support each problem with evidence from the client meeting (numbers, quotes, examples). End with a one-sentence proposed solution.*

ID	Current process / problem	Impact on client	Evidence (source)
P1	Outages are reported by phone and call volume spikes in big outages	Long wait times	Client interview, dd/mm
P2	Reports lack exact location	Extra call-backs	number or estimate from client
P3	No updates after reporting	Repeat calls	source

Table 1: Summary of client problems

▸ *Proposed solution (1–2 sentences): "A Flutter mobile app that … using AI to …".*

3. Project Objectives & Success Metrics

PRD section: Goals & Success Metrics **Rubric link:** Objectives well defined and prioritised

‣ *3–4 objectives covering the lifecycle (study → design/develop → test/evaluate). Each must be measurable and linked to a problem (P-ID).*

No.	Objective	Solves	Success metric (how we measure)	Priority
O1	Study the current outage reporting process with the client	P1-P3	Requirements confirmed by Week 9	*High*
O2	Develop an Android app for reporting and tracking outages	P1-P3	All Must-have FRs pass acceptance tests	*High*
O3	Integrate AI photo classification	P2	At least 80% accuracy on a 50-photo test set	*High*
O4	Test and evaluate with the client	*All*	At least 50 tests, at least 60% coverage, UAT sign-off	*Medium*

Table 2: Project objectives and success metrics

4. Project Perspective

PRD section: Target Users & System Context **Rubric link:** Information obtained and analysed

4.1 Target Users

‣ *List each user type (persona). Keep to 2–4 user types.*

User type	Description	Main needs / goals	Device / tech level
e.g. Customer			*Android phone, basic*
e.g. Admin (client)			

Table 3: Target users

4.2 System Context and Architecture

‣ *Show how the app fits into the client's environment: is it new, replacing, or extending an existing system? Include a high-level architecture diagram: Flutter app ↔ backend (Firebase / Supabase / REST) ↔ AI API ↔ external services (maps, payment, etc.). Explain the diagram in 1 paragraph.*

[Insert Figure 1: High-level system architecture]

5. Project Scope

PRD section: Scope, Features & Requirements **Rubric link:** Design recommendations supported

5.1 In Scope / Out of Scope

‣ *Draw a clear boundary. Out-of-scope items prevent scope creep and are just as important as in-scope items.*

In scope (will deliver this semester)	Out of scope (will not deliver)
e.g. Android app for customers and admin	*e.g. Online payment gateway*

Table 4: Scope boundary

5.2 Features and User Stories

‣ Group features into modules. Write each as a user story. Prioritise with MoSCoW (Must / Should / Could / Won't). These FR-IDs carry forward into the SRS and GitHub Issues.

FR-ID	Module	User story (As a … I want … so that …)	Priority
FR-01	*User Management*	*As a customer, I want to register and log in so that my orders are saved.*	*Must*
FR-02			
FR-03			
FR-AI-01	*AI Feature*		*Must*

Table 5: Features and user stories

[Insert Figure 2: System modules diagram]

5.3 AI Feature

‣ At least one AI feature is mandatory. In 1 short paragraph plus the table, explain what it does and why it adds real value (not a gimmick).

Item	Description
AI capability	e.g. chatbot, OCR, recommendation, classification, prediction
User value	Which problem (P-ID) it solves and for which user
Input → Output	What data goes in and what the user sees
AI service / API	e.g. Gemini, Claude, OpenAI, on-device ML Kit
Constraints	API key protection, cost, latency, offline fallback, data privacy

Table 6: AI feature summary

5.4 Non-Functional Requirements and Constraints

NFR-ID	Category	Requirement / constraint	Target
NFR-01	Performance	Outage report submissions and photo uploads must send quickly over mobile data.	Response confirmation in < 2.0 s on 3G/4G.
NFR-02	Security	API keys, database credentials, and passwords must never be visible in public source code.	0 plain-text secrets in the repository.
NFR-03	Usability	App screens must be easy to read and navigate during power cuts.	100% Material Design 3 compliance; report completed in ≤ 3 taps.
NFR-04	Reliability / AI	If the AI service fails or drops connection, the user must still be able to select a category manually.	Display manual dropdown fallback within 3.0 s.
NFR-05	Compatibility	The app must run on standard consumer phones.	Supports Android 8.0+ and iOS 12+.

Table 7: Non-functional requirements and constraints

System Constraints:

1. **Time Constraint:** Must be delivered by the end of the 15-week semester (2 January 2027).
2. **Resource Constraint:** Relies on free-tier cloud tools (Firebase, free AI API credits).
3. **Data Privacy:** Customer data and photos must comply with basic UNIMAS academic ethics and Sarawak Energy guidelines.

5.5 System Environment

Layer	Technology / tool	Justification
Mobile app	Flutter (Dart 3), Material 3	Course requirement; cross-platform
State management	e.g. Provider / Riverpod	
Backend / database	e.g. Firebase Auth + Cloud Firestore	
AI service		
Version control & CI	GitHub, GitHub Actions	Course requirement
Target devices	e.g. Android 8.0+ phones	

Table 8: System environment

6. Task Allocation & Team Contribution

PRD section: Team & Stakeholders (RACI) **Rubric link:** Project breakdown; team work

▸ *One row per member. Every member must have at least one technical task. Assign a TSPi role and a feature ownership (FR-IDs) to each.*

Member	TSPi role	Owned features (FR-IDs)	Other responsibilities / contribution
	Team Leader		*e.g. weekly meetings, WEEK reports*
	Development Manager		*e.g. architecture, code review*
	Planning Manager		*e.g. TASK / SCHEDULE, GitHub board*
Fabian Anak Sudin	Quality/Process Manager + Support Manager	**FR-04** (View Report Status & Updates)	GitHub repository setup, CI/CD pipeline script, LOGAI tracking log, LOGTEST execution, automated test coverage setup.
	Client Liaison / Developer		*e.g. client meetings, UAT*

Table 9: Task allocation and contribution

7. Team Goals

PRD section: Team Working Agreement **Rubric link:** Team work

▸ Maximum 4 goals. Each needs a measure so it can be checked at the weekly meeting (refer to the TSPi lecture on team goals).

No.	Team goal	Measure
G1	*e.g. Produce a quality product*	*e.g. All Must-have FRs pass; CI green on main*
G2	*e.g. Run a productive, well-managed project*	*e.g. Every milestone delivered on time*
G3	*e.g. Every member contributes fairly*	*e.g. Each member merges ≥1 PR per sprint*
G4	*e.g. Keep the client engaged*	*e.g. Client meeting at least every 2 weeks*

Table 10: Team goals

8. Expected Results

PRD section: Release Criteria / Deliverables **Rubric link:** Decisions supported by information

8.1 Working System

The main deliverable is the **OutageWatchSarawak** mobile app developed in Flutter for electricity customers, field technicians, and control-room staff. The app allows customers to submit power cut reports with photo proof and GPS locations, uses AI to automatically suggest the incident type (such as fallen lines or transformer damage), and provides real-time restoration updates. System acceptance requires a functional release APK (v1.0.0), a passing GitHub Actions build, at least 50 passing test cases with 60% code coverage, and a signed client acceptance form.

8.2 Documentation and Artefacts

Deliverable	Submitted with
Project Proposal (original)	Proposal Submission (16 October 2026)
Revised Proposal and Software Requirements Specification (SRS)	Initial Report (Week 9)
Runnable prototype (4–5 screens), tagged release	Demo 1 (Week 9)
Software Design Specification (SDS), test plan and results (LOGTEST), user manual, maintenance plan, client acceptance form	Final Report (Week 13)
Release APK, demo video (5–10 min), presentation slides	Final Demo 2 (Week 14)
GitHub repository and complete project file	e-Archive (Week 15)
TSPi forms (LOGT, LOGD, LOGAI, SCHEDULE, SUMP, SUMQ, SUMS, SUMTASK, WEEK, PEER) and meeting minutes	Ongoing; compiled in Final Report

Table 11: Deliverables

9. Project Planning

PRD section: Milestones, Timeline & Risks **Rubric link:** Methodology to be followed

9.1 Methodology

‣ One paragraph: state the Agile approach (e.g. Scrum with 2-week sprints), why it suits this client, and how it runs with TSPi roles and the GitHub workflow (issues → branch → PR review → CI). Include a small framework figure if helpful.

9.2 Submission Schedule

‣ Fixed course deadlines. Plan your sprints backwards from these dates.

Phase	Deliverable	Week	Due
Phase 1: Planning & Development	Proposal Submission	4	9 Oct 2026
	Proposal Presentation	5	19 – 23 Oct 2026
	Initial Report	9	16 Nov 2026
	Demo 1	9	16 – 20 Nov 2026
Phase 2: Finalisation & Closure	Final Report	13	24 Dec 2026
	Final Demo 2	14	28 – 31 Dec 2026
	e-Archive	15	Before 10 Jan 2027

Table 12: Submission schedule

9.3 Development Milestones (GitHub)

Milestone	Week	Key output	Status
M1 – Scope & Requirements	4 – 5	GitHub repo, issue board, CI file; proposal and initial SRS in /docs	
M2 – Design & Prototype	9	SDS draft, 4–5 screen prototype, green CI, v0.2.0 tag	
M3 – Product & Verification	13	≥50 tests, ≥60% coverage, AI feature working	
M4 – Final Release & Acceptance	15	Release APK v1.0.0, client sign-off, e-Archive	

Table 13: Development milestones

‣ Insert a Gantt chart (Figure 3) from Week 3 to Week 15 showing sprints, the seven submission dates and the four milestones. It should match your TSPi TASK and SCHEDULE worksheets.

[Insert Figure 3: Gantt chart]

9.4 Risks and Assumptions

ID	Risk / assumption	Likelihood	Impact	Mitigation
R1	Client is busy and unavailable for regular meetings.	Medium	High	Fix bi-weekly meeting slots early and use online/email signoffs.
R2	AI fails on dark or blurry photos or reaches daily API limits.	Medium	Medium	Provide a simple manual dropdown menu so users can pick the category themselves.
R3	Team members struggle with complex Flutter code.	High	High	Require peer Pull Request reviews before merging code to main.

ID	Risk / assumption	Likelihood	Impact	Mitigation
R4	Scope becomes too big near the deadline.	Medium	High	Keep features simple and prioritises core functional needs first.

Table 14: Risks and assumptions

Assumptions:

1. Users have basic mobile data (3G/4G) or location services enabled when sending a report.
2. User phones meet the minimum standard of Android 8.0 or iOS 12.
3. Client representatives from Sarawak Energy remain available for requirement feedback and testing.

10. References

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- Jali, N. B. (2026). *Invitation to participate in a stakeholder interview: OutageWatch Sarawak*. Faculty of Computer Science and Information Technology, Universiti Malaysia Sarawak.[cite: 55]

Appendix A: Client Meeting Evidence

- *Meeting minutes summary (date, attendees, key decisions) and, if allowed, a photo with the client.*

Appendix B: Revision History

Version	Date	Changes (e.g. after proposal presentation feedback)	By
1.0	*16/10/2026*	*Initial submission of Project Proposal (PRD) for OutageWatchSarawak.*	*Group Biscuding (Dayrill Muller, Fabian, Maccoy, Bricolen Benjamin)*